Cloud Computing

UE23CS351B

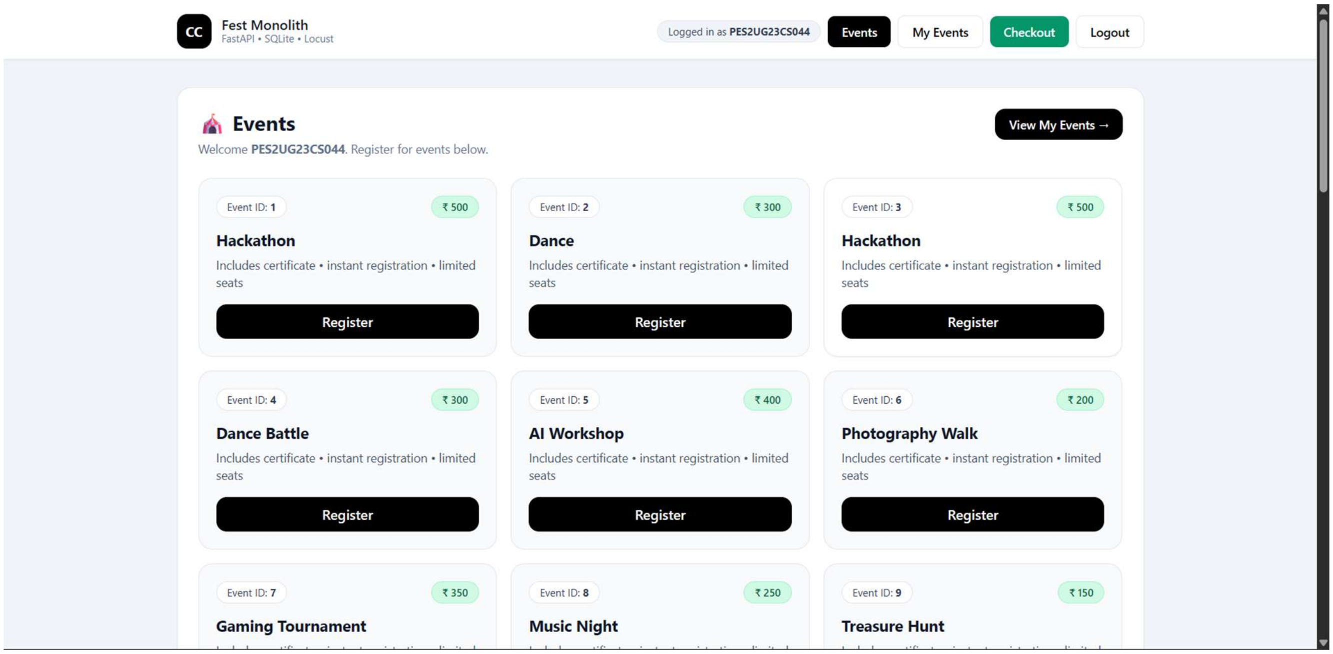
LAB 2- MONOLITHIC ARCHITECTURE

GitHub Link : https://github.com/akshayag2005/CLOUD-COMPUTING.git

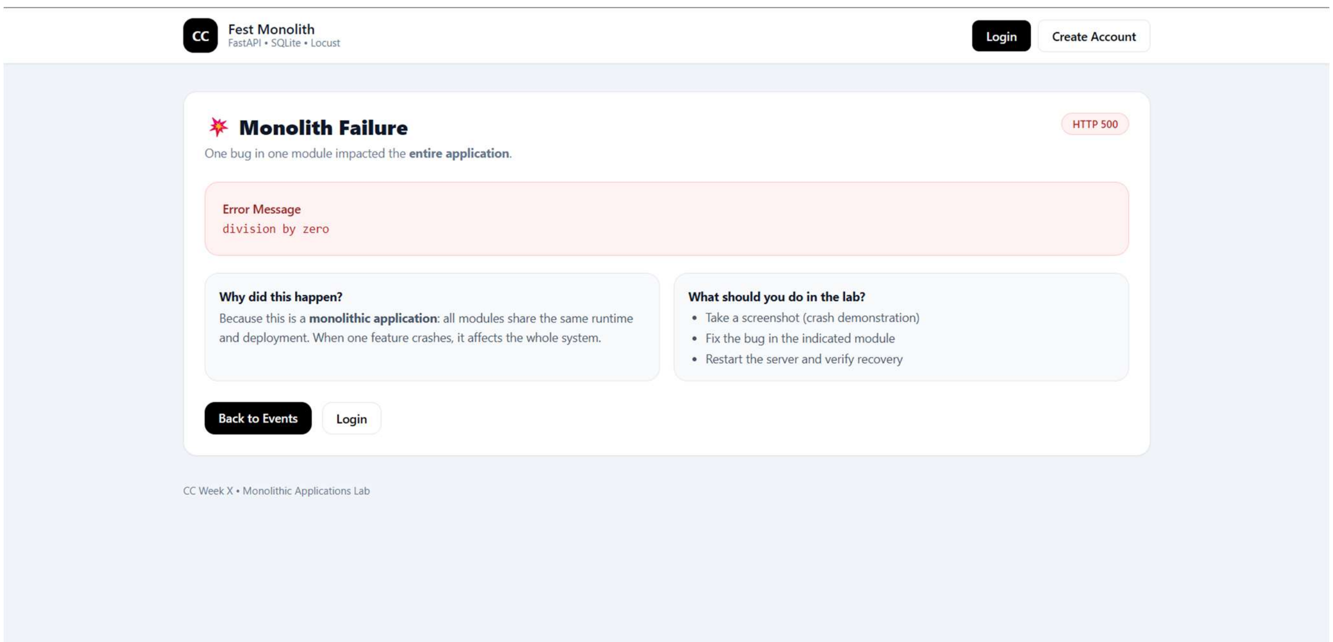
DATE: 20/01/2026

NAME: AKSHAY
SRN: PES2UG23CS044
SECTION: A

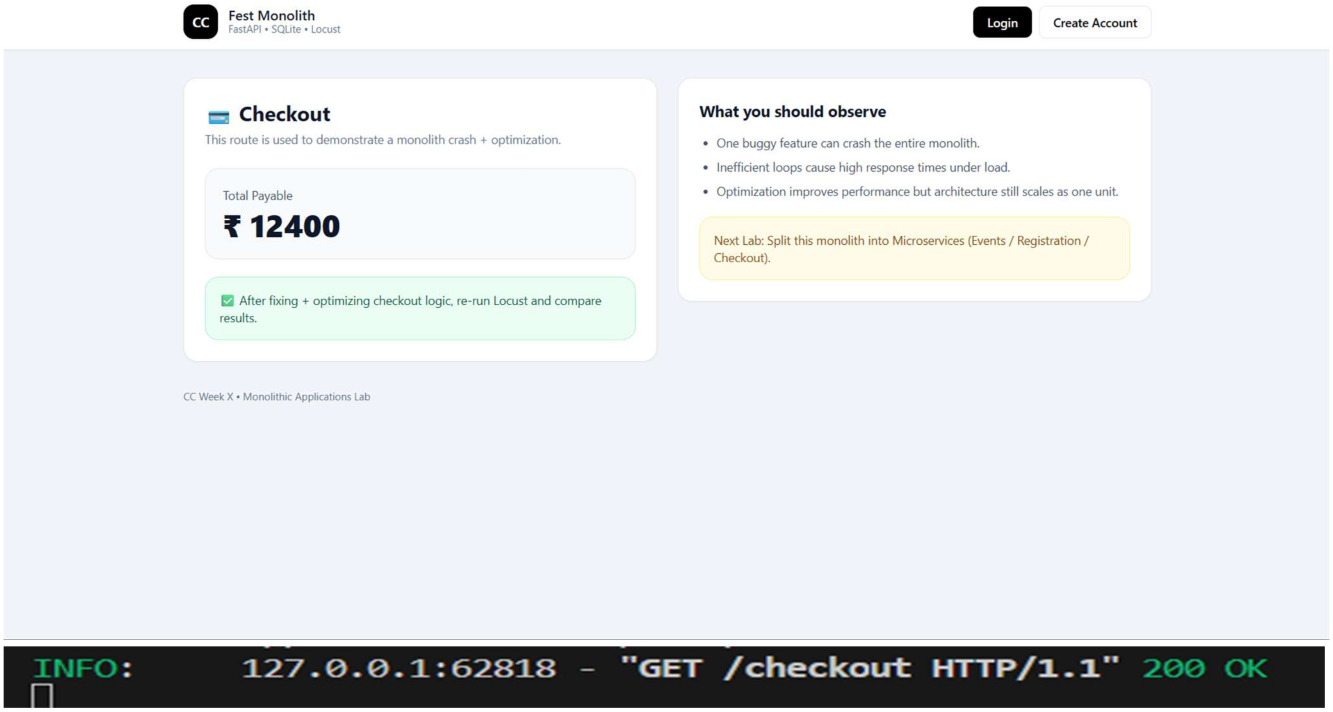
1. ## Screenshot-SS1 (Events page loaded)



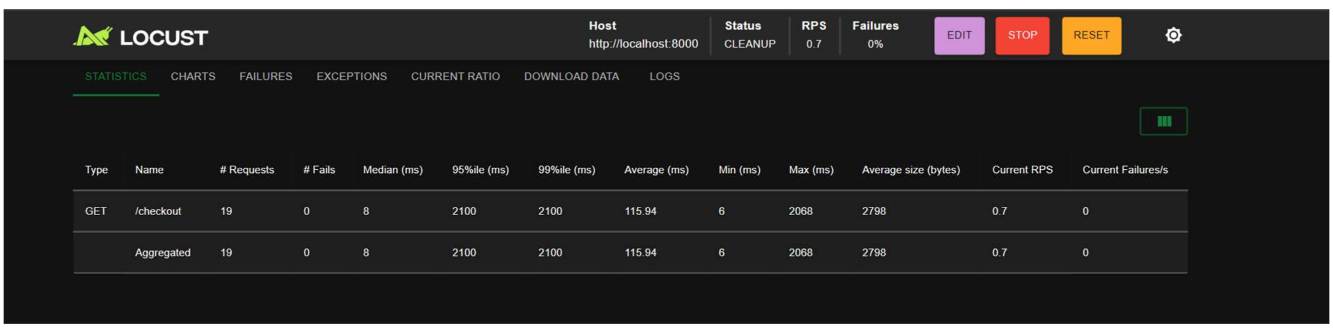
2. ## Screenshot-SS2

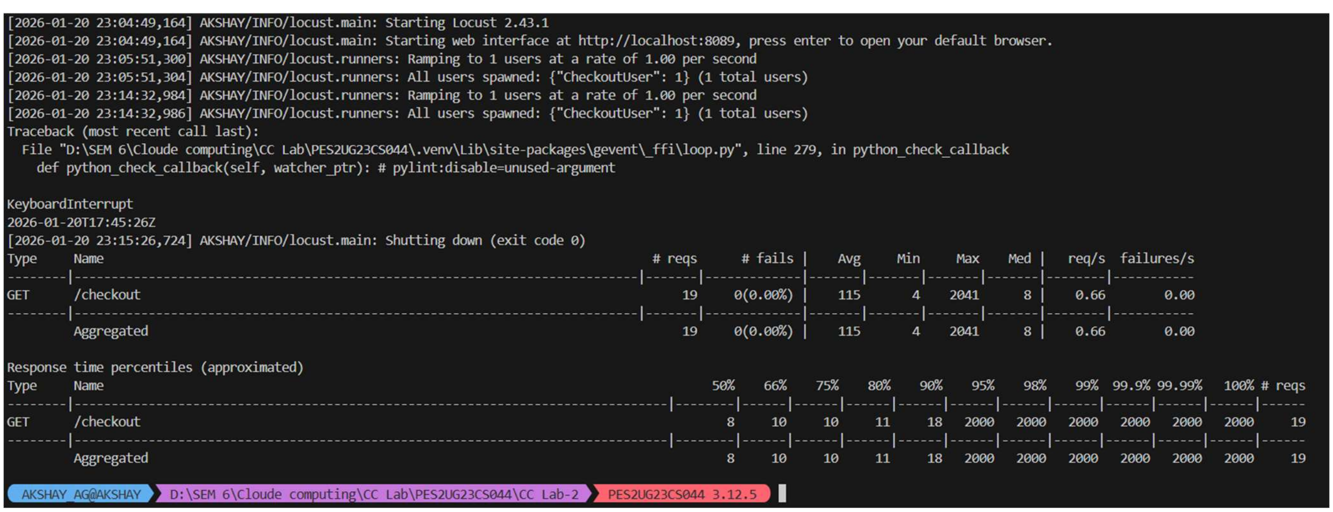


3. Screenshot-SS3

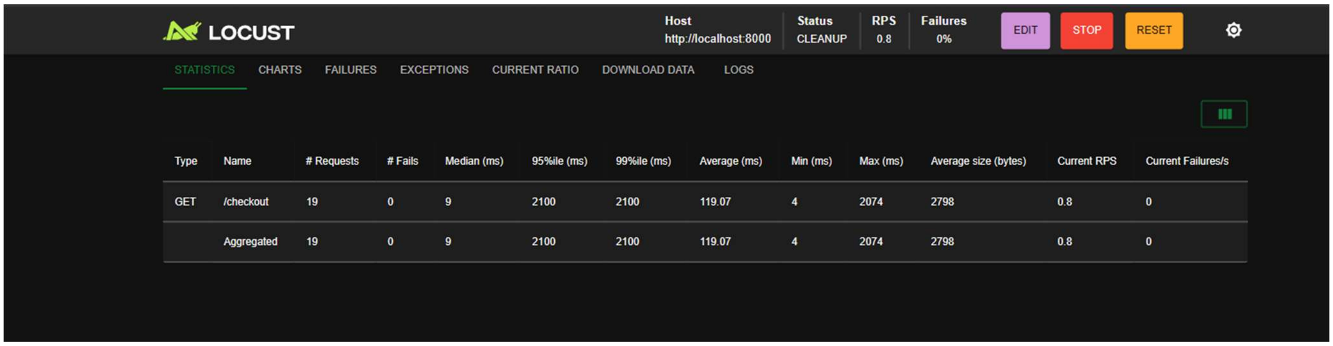


4. Screenshot-SS4





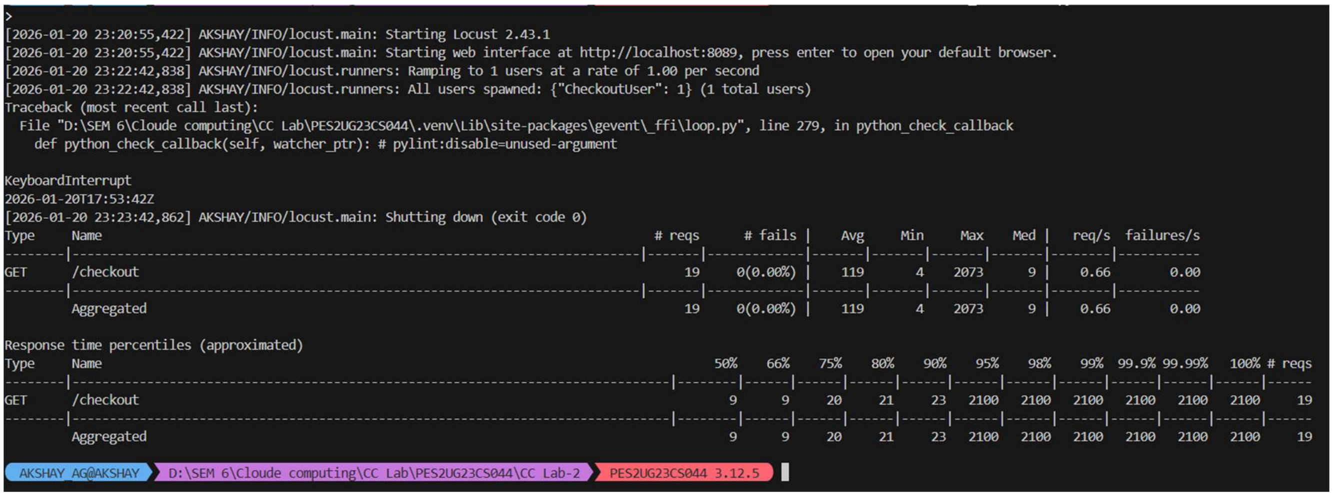
5. Screenshot-SS5



LOCUST | Host: http://localhost:8000 | Status: CLEANUP | RPS: 0.8 | Failures: 0% | EDIT | STOP | RESET

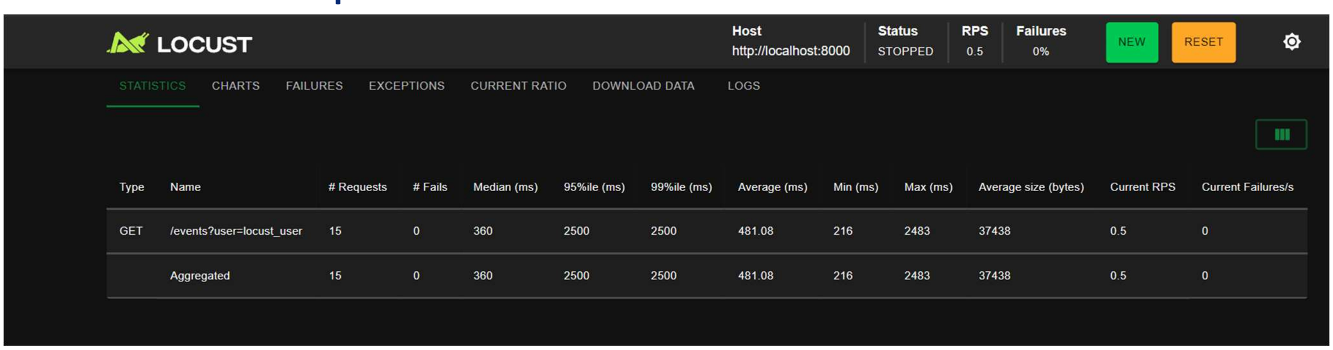
STATISTICS | CHARTS | FAILURES | EXCEPTIONS | CURRENT RATIO | DOWNLOAD DATA | LOGS

Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	Current RPS	Current Failures/s
GET	/checkout	19	0	9	2100	2100	119.07	4	2074	2798	0.8	0
	Aggregated	19	0	9	2100	2100	119.07	4	2074	2798	0.8	0



```
>
[2026-01-20 23:20:55,422] AKSHAY/INFO/locust.main: Starting Locust 2.43.1
[2026-01-20 23:20:55,422] AKSHAY/INFO/locust.main: Starting web interface at http://localhost:8089, press enter to open your default browser.
[2026-01-20 23:22:42,838] AKSHAY/INFO/locust.runners: Ramping to 1 users at a rate of 1.00 per second
[2026-01-20 23:22:42,838] AKSHAY/INFO/locust.runners: All users spawned: {"CheckoutUser": 1} (1 total users)
Traceback (most recent call last):
  File "D:\SEM 6\Cloude computing\CC Lab\PES2UG23CS044\.venv\Lib\site-packages\gevent\_ffi\loop.py", line 279, in python_check_callback
    def python_check_callback(self, watcher_ptr): # pylint:disable=unused-argument

KeyboardInterrupt
2026-01-20T17:53:42Z
[2026-01-20 23:23:42,862] AKSHAY/INFO/locust.main: Shutting down (exit code 0)
Type     Name                                                                          # reqs      # fails |    Avg     Min     Max    Med |   req/s  failures/s
--------|----------------------------------------------------------------------------|-------|-------------|-------|-------|-------|-------|--------|-----------
GET      /checkout                                                                         19     0(0.00%) |    119       4    2073      9 |    0.66        0.00
--------|----------------------------------------------------------------------------|-------|-------------|-------|-------|-------|-------|--------|-----------
         Aggregated                                                                        19     0(0.00%) |    119       4    2073      9 |    0.66        0.00

Response time percentiles (approximated)
Type     Name                                                                                  50%    66%    75%    80%    90%    95%    98%    99%  99.9% 99.99%   100% # reqs
--------|--------------------------------------------------------------------------------|--------|------|------|------|------|------|------|------|------|------|------|------
GET      /checkout                                                                               9      9     20     21     23   2100   2100   2100   2100   2100   2100     19
--------|--------------------------------------------------------------------------------|--------|------|------|------|------|------|------|------|------|------|------|------
         Aggregated                                                                              9      9     20     21     23   2100   2100   2100   2100   2100   2100     19

AKSHAY_AG@AKSHAY  D:\SEM 6\Cloude computing\CC Lab\PES2UG23CS044\CC Lab-2  PES2UG23CS044 3.12.5
```

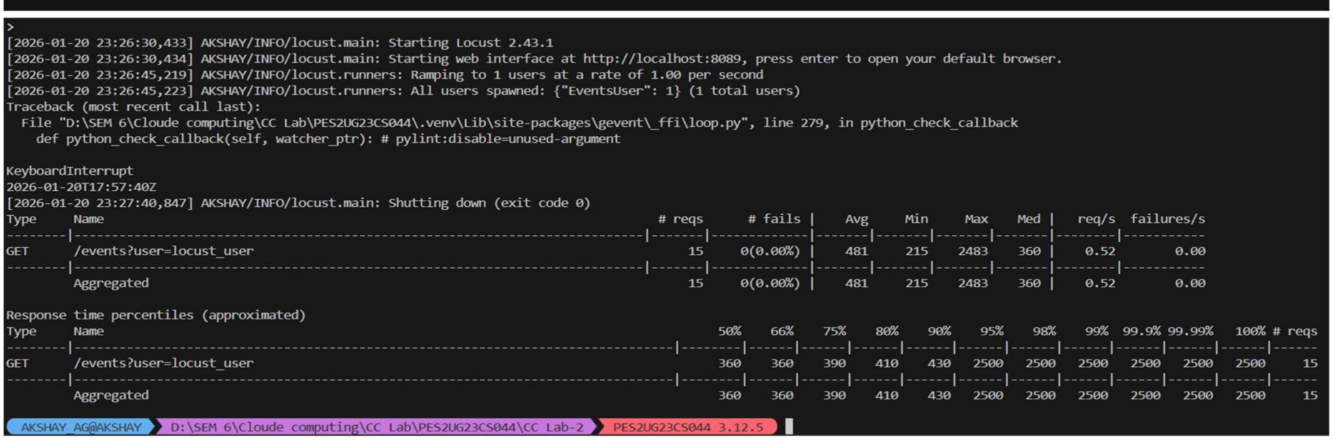
6. Screenshot BEFORE optimization-SS6



LOCUST | Host: http://localhost:8000 | Status: STOPPED | RPS: 0.5 | Failures: 0% | NEW | RESET

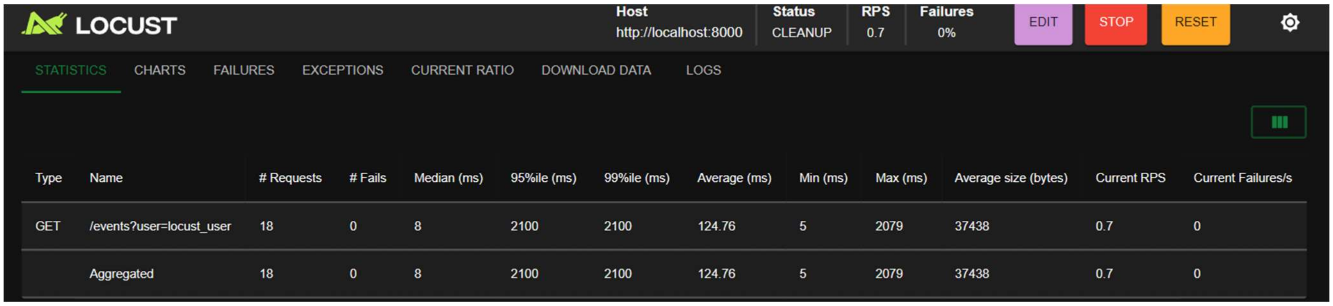
STATISTICS | CHARTS | FAILURES | EXCEPTIONS | CURRENT RATIO | DOWNLOAD DATA | LOGS

Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	Current RPS	Current Failures/s
GET	/events?user=locust_user	15	0	360	2500	2500	481.08	216	2483	37438	0.5	0
	Aggregated	15	0	360	2500	2500	481.08	216	2483	37438	0.5	0

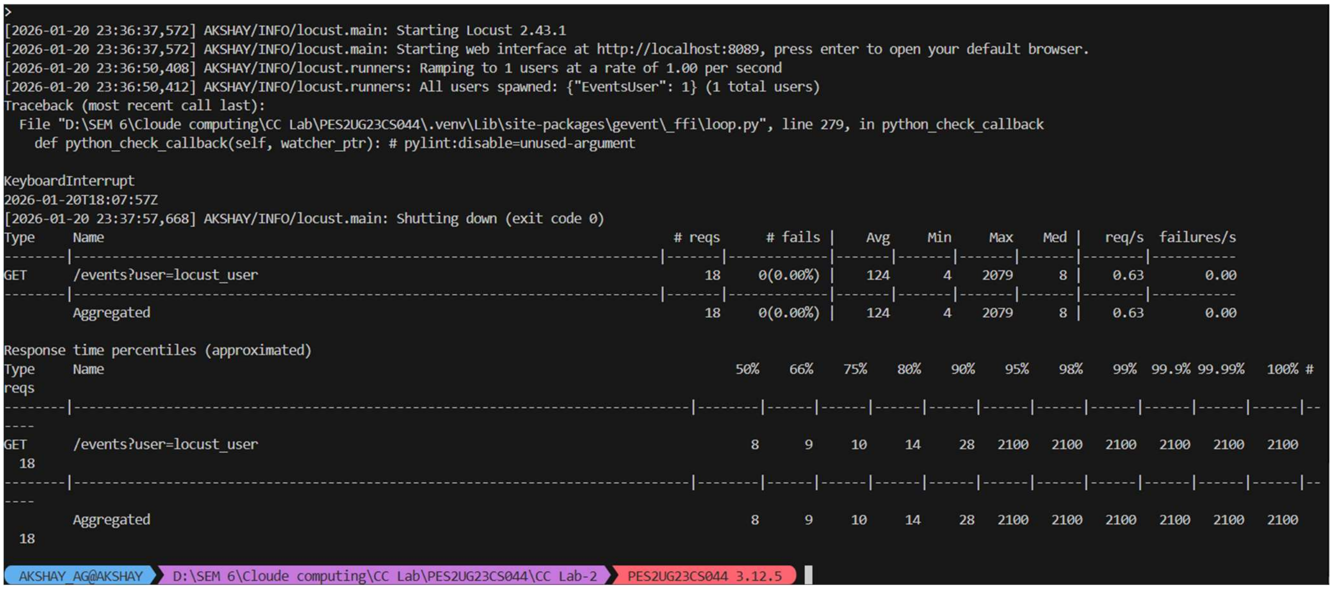


```
>
[2026-01-20 23:26:30,433] AKSHAY/INFO/locust.main: Starting Locust 2.43.1
[2026-01-20 23:26:30,434] AKSHAY/INFO/locust.main: Starting web interface at http://localhost:8089, press enter to open your default browser.
[2026-01-20 23:26:45,219] AKSHAY/INFO/locust.runners: Ramping to 1 users at a rate of 1.00 per second
[2026-01-20 23:26:45,223] AKSHAY/INFO/locust.runners: All users spawned: {"EventsUser": 1} (1 total users)
Traceback (most recent call last):
  File "D:\SEM 6\Cloude computing\CC Lab\PES2UG23CS044\.venv\Lib\site-packages\gevent\_ffi\loop.py", line 279, in python_check_callback
    def python_check_callback(self, watcher_ptr): # pylint:disable=unused-argument

KeyboardInterrupt
2026-01-20T17:57:40Z
[2026-01-20 23:27:40,847] AKSHAY/INFO/locust.main: Shutting down (exit code 0)
Type     Name                                                                          # reqs      # fails |    Avg     Min     Max    Med |   req/s  failures/s
--------|----------------------------------------------------------------------------|-------|-------------|-------|-------|-------|-------|--------|-----------
GET      /events?user=locust_user                                                          15     0(0.00%) |    481     215    2483    360 |    0.52        0.00
--------|----------------------------------------------------------------------------|-------|-------------|-------|-------|-------|-------|--------|-----------
         Aggregated                                                                        15     0(0.00%) |    481     215    2483    360 |    0.52        0.00

Response time percentiles (approximated)
Type     Name                                                                                  50%    66%    75%    80%    90%    95%    98%    99%  99.9% 99.99%   100% # reqs
--------|--------------------------------------------------------------------------------|--------|------|------|------|------|------|------|------|------|------|------|------
GET      /events?user=locust_user                                                              360    360    390    410    430   2500   2500   2500   2500   2500   2500     15
--------|--------------------------------------------------------------------------------|--------|------|------|------|------|------|------|------|------|------|------|------
         Aggregated                                                                            360    360    390    410    430   2500   2500   2500   2500   2500   2500     15

AKSHAY_AG@AKSHAY  D:\SEM 6\Cloude computing\CC Lab\PES2UG23CS044\CC Lab-2  PES2UG23CS044 3.12.5
```

7. Screenshot AFTER optimization-SS7

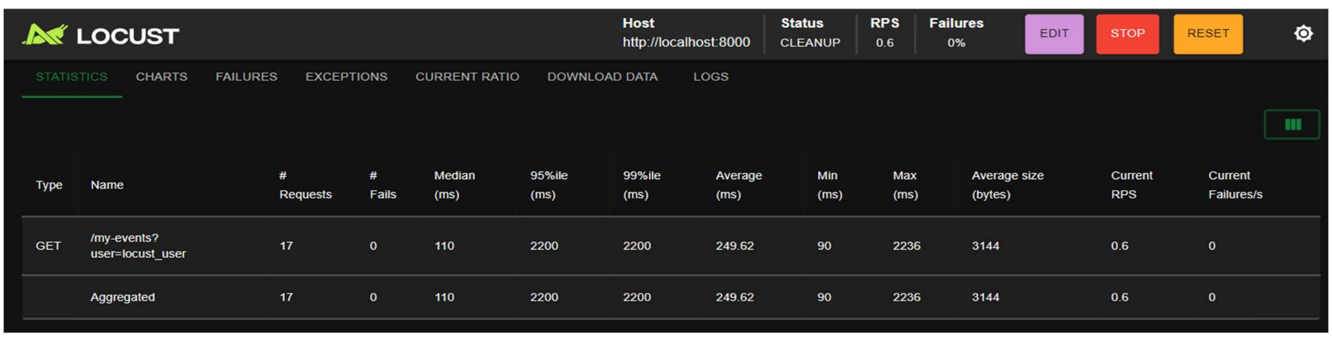


Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	Current RPS	Current Failures/s
GET	/events?user=locust_user	18	0	8	2100	2100	124.76	5	2079	37438	0.7	0
	Aggregated	18	0	8	2100	2100	124.76	5	2079	37438	0.7	0

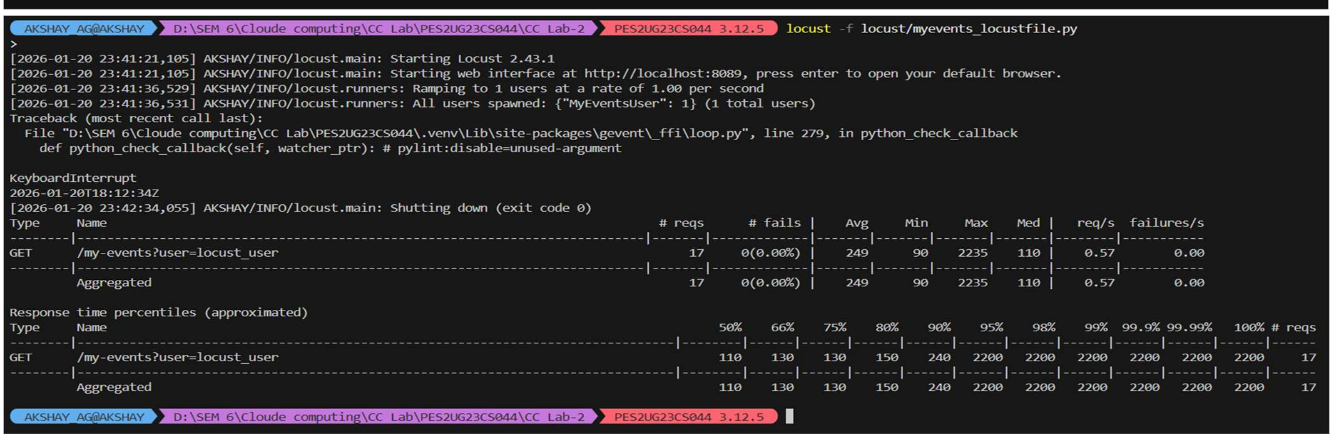


```
>
[2026-01-20 23:36:37,572] AKSHAY/INFO/locust.main: Starting Locust 2.43.1
[2026-01-20 23:36:37,572] AKSHAY/INFO/locust.main: Starting web interface at http://localhost:8089, press enter to open your default browser.
[2026-01-20 23:36:50,408] AKSHAY/INFO/locust.runners: Ramping to 1 users at a rate of 1.00 per second
[2026-01-20 23:36:50,412] AKSHAY/INFO/locust.runners: All users spawned: {"EventsUser": 1} (1 total users)
Traceback (most recent call last):
  File "D:\SEM 6\Cloude computing\CC Lab\PES2UG23CS044\.venv\Lib\site-packages\gevent\_ffi\loop.py", line 279, in python_check_callback
    def python_check_callback(self, watcher_ptr): # pylint:disable=unused-argument

KeyboardInterrupt
2026-01-20T18:07:57Z
[2026-01-20 23:37:57,668] AKSHAY/INFO/locust.main: Shutting down (exit code 0)
Type     Name                                                                          # reqs      # fails |    Avg     Min     Max    Med |   req/s  failures/s
--------|----------------------------------------------------------------------------|-------|-------------|-------|-------|-------|-------|--------|-----------
GET      /events?user=locust_user                                                          18     0(0.00%) |    124       4    2079      8 |    0.63        0.00
--------|----------------------------------------------------------------------------|-------|-------------|-------|-------|-------|-------|--------|-----------
         Aggregated                                                                        18     0(0.00%) |    124       4    2079      8 |    0.63        0.00

Response time percentiles (approximated)
Type     Name                                                                                  50%    66%    75%    80%    90%    95%    98%    99%  99.9% 99.99%   100% #
reqs
--------|--------------------------------------------------------------------------------|--------|------|------|------|------|------|------|------|------|------|------|--
----
GET      /events?user=locust_user                                                                8      9     10     14     28   2100   2100   2100   2100   2100   2100
  18
--------|--------------------------------------------------------------------------------|--------|------|------|------|------|------|------|------|------|------|------|--
----
         Aggregated                                                                              8      9     10     14     28   2100   2100   2100   2100   2100   2100
  18
AKSHAY_AG@AKSHAY  D:\SEM 6\Cloude computing\CC Lab\PES2UG23CS044\CC Lab-2  PES2UG23CS044 3.12.5
```

8. Screenshot BEFORE optimization-SS8

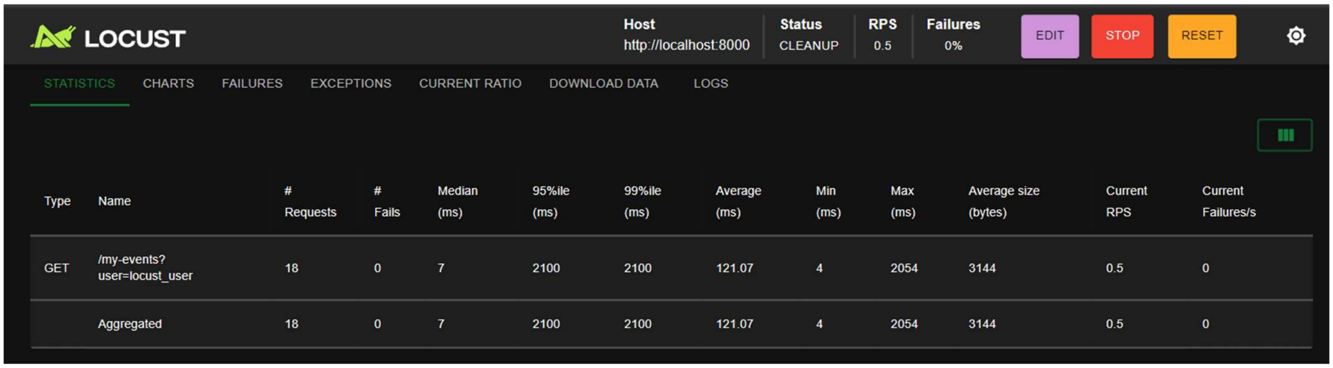


Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	Current RPS	Current Failures/s
GET	/my-events?user=locust_user	17	0	110	2200	2200	249.62	90	2236	3144	0.6	0
	Aggregated	17	0	110	2200	2200	249.62	90	2236	3144	0.6	0

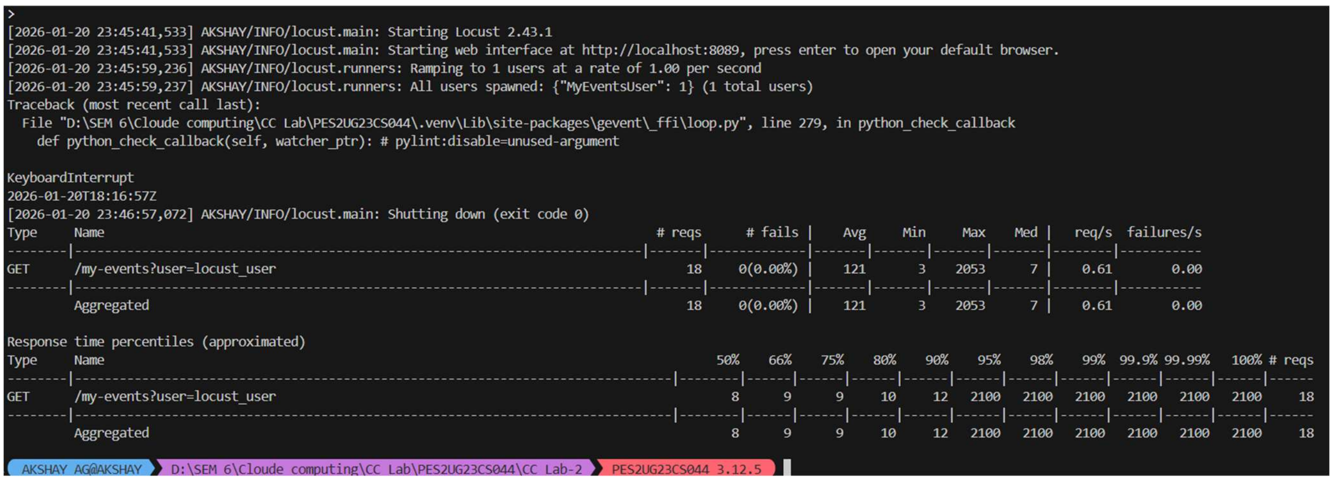


```
AKSHAY_AG@AKSHAY  D:\SEM 6\Cloude computing\CC Lab\PES2UG23CS044\CC Lab-2  PES2UG23CS044 3.12.5  locust -f locust/myevents_locustfile.py
>
[2026-01-20 23:41:21,105] AKSHAY/INFO/locust.main: Starting Locust 2.43.1
[2026-01-20 23:41:21,105] AKSHAY/INFO/locust.main: Starting web interface at http://localhost:8089, press enter to open your default browser.
[2026-01-20 23:41:36,529] AKSHAY/INFO/locust.runners: Ramping to 1 users at a rate of 1.00 per second
[2026-01-20 23:41:36,531] AKSHAY/INFO/locust.runners: All users spawned: {"MyEventsUser": 1} (1 total users)
Traceback (most recent call last):
  File "D:\SEM 6\Cloude computing\CC Lab\PES2UG23CS044\.venv\Lib\site-packages\gevent\_ffi\loop.py", line 279, in python_check_callback
    def python_check_callback(self, watcher_ptr): # pylint:disable=unused-argument

KeyboardInterrupt
2026-01-20T18:12:34Z
[2026-01-20 23:42:34,055] AKSHAY/INFO/locust.main: Shutting down (exit code 0)
Type     Name                                                                          # reqs      # fails |    Avg     Min     Max    Med |   req/s  failures/s
--------|----------------------------------------------------------------------------|-------|-------------|-------|-------|-------|-------|--------|-----------
GET      /my-events?user=locust_user                                                       17     0(0.00%) |    249      90    2235    110 |    0.57        0.00
--------|----------------------------------------------------------------------------|-------|-------------|-------|-------|-------|-------|--------|-----------
         Aggregated                                                                        17     0(0.00%) |    249      90    2235    110 |    0.57        0.00

Response time percentiles (approximated)
Type     Name                                                                                  50%    66%    75%    80%    90%    95%    98%    99%  99.9% 99.99%   100% # reqs
--------|--------------------------------------------------------------------------------|--------|------|------|------|------|------|------|------|------|------|------|------
GET      /my-events?user=locust_user                                                           110    130    130    150    240   2200   2200   2200   2200   2200   2200     17
--------|--------------------------------------------------------------------------------|--------|------|------|------|------|------|------|------|------|------|------|------
         Aggregated                                                                            110    130    130    150    240   2200   2200   2200   2200   2200   2200     17
AKSHAY_AG@AKSHAY  D:\SEM 6\Cloude computing\CC Lab\PES2UG23CS044\CC Lab-2  PES2UG23CS044 3.12.5
```

9. Screenshot AFTER optimization-SS9



Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	Current RPS	Current Failures/s
GET	/my-events?user=locust_user	18	0	7	2100	2100	121.07	4	2054	3144	0.5	0
	Aggregated	18	0	7	2100	2100	121.07	4	2054	3144	0.5	0



```
>
[2026-01-20 23:45:41,533] AKSHAY/INFO/locust.main: Starting Locust 2.43.1
[2026-01-20 23:45:41,533] AKSHAY/INFO/locust.main: Starting web interface at http://localhost:8089, press enter to open your default browser.
[2026-01-20 23:45:59,236] AKSHAY/INFO/locust.runners: Ramping to 1 users at a rate of 1.00 per second
[2026-01-20 23:45:59,237] AKSHAY/INFO/locust.runners: All users spawned: {"MyEventsUser": 1} (1 total users)
Traceback (most recent call last):
  File "D:\SEM 6\Cloude computing\CC Lab\PES2UG23CS044\.venv\Lib\site-packages\gevent\_ffi\loop.py", line 279, in python_check_callback
    def python_check_callback(self, watcher_ptr): # pylint:disable=unused-argument

KeyboardInterrupt
2026-01-20T18:16:57Z
[2026-01-20 23:46:57,072] AKSHAY/INFO/locust.main: Shutting down (exit code 0)
Type     Name                                                                          # reqs      # fails |    Avg     Min     Max    Med |   req/s  failures/s
--------|----------------------------------------------------------------------------|-------|-------------|-------|-------|-------|-------|--------|-----------
GET      /my-events?user=locust_user                                                       18     0(0.00%) |    121       3    2053      7 |    0.61        0.00
--------|----------------------------------------------------------------------------|-------|-------------|-------|-------|-------|-------|--------|-----------
         Aggregated                                                                        18     0(0.00%) |    121       3    2053      7 |    0.61        0.00

Response time percentiles (approximated)
Type     Name                                                                                  50%    66%    75%    80%    90%    95%    98%    99%  99.9% 99.99%   100% # reqs
--------|--------------------------------------------------------------------------------|--------|------|------|------|------|------|------|------|------|------|------|------
GET      /my-events?user=locust_user                                                             8      9      9     10     12   2100   2100   2100   2100   2100   2100     18
--------|--------------------------------------------------------------------------------|--------|------|------|------|------|------|------|------|------|------|------|------
         Aggregated                                                                              8      9      9     10     12   2100   2100   2100   2100   2100   2100     18

 AKSHAY AG@AKSHAY  D:\SEM 6\Cloude computing\CC Lab\PES2UG23CS044\CC Lab-2  PES2UG23CS044 3.12.5
```

• Route 1: /events

1. **What was the bottleneck?**
   The /events route contained inefficient logic where event data was processed repeatedly and unnecessary looping was performed while constructing the response. This increased CPU usage for every request.
2. **What change did you make?**
   The code was optimized by removing redundant loops and processing the event data in a single pass. Only the required fields were accessed directly instead of repeatedly recalculating values.
3. **Why did the performance improve?**
   By reducing unnecessary computations and iterations, the CPU workload per request decreased. This resulted in lower response time and improved overall performance during load testing.

• Route 2: /my-events

1. **What was the bottleneck?**
   The /my-events route performed repeated filtering and processing of the same dataset for every request, leading to inefficient use of CPU resources.
2. **What change did you make?**
   The logic was optimized by simplifying the filtering process and avoiding repeated scans of the data. The required information was computed more efficiently in a single operation.
3. **Why did the performance improve?**
   Optimizing the filtering logic reduced redundant processing, which lowered execution time per request. As a result, response time improved when tested using Locust.